Q1. House Price Prediction A real-estate company builds a Linear Regression model using Area, Bedrooms, Bathrooms, and Age to predict house prices.

The residual plot shows a curved pattern, and prediction errors increase as predicted prices increase.

```
#1.. Identify the assumptions that may be violated.
Linearity: The curved pattern in the residual plot indicates that the relati
 between independent variables and house price may not be purely linear.
Homoscedasticity: The increasing prediction errors indicate that the varianc
 residuals is not constant. This problem is known as heteroscedasticity.
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#2.. Explain why a curved residual pattern is important.
A curved residual pattern is important because it indicates that the Linear
properly capture the relationship between independent and dependent variable
- Residuals should be randomly distributed around the zero line.
- A curved pattern indicates a non-linear relationship.
- It suggests that the linearity assumption of regression is violated.
- The model may produce inaccurate predictions because it cannot capture the
- Polynomial regression or feature transformation can be used to handle this
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# 3..Explain why increasing residual variance can be a problem.
Increasing residual variance is a problem because it indicates Heteroscedast
error variance is not constant. It makes regression estimates and standard e
which can lead to incorrect statistical conclusions.
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# 4..Suggest appropriate diagnostic plots or statistical tests.
Residual vs Fitted Plot: Used to identify non-linearity and unequal residual
Breusch-Pagan Test: Used to statistically detect heteroscedasticity.
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# 5..Recommend possible solutions for each problem.
For Non-linearity: Use Polynomial Regression or apply suitable feature trans
For Heteroscedasticity: Apply log transformation to the target variable or u
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Sales Forecasting A retail company uses Linear Regression to predict daily sales. The residuals show a repeating time pattern, and the Durbin-Watson statistic is 0.8.

Perform the following tasks:

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# 1..Identify the likely regression assumption violation.
The likely violated assumption is Independence of Errors (No Autocorrelation
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# 2..Explain what a Durbin-Watson statistic below 2 can indicate.
A Durbin-Watson statistic below 2 indicates positive autocorrelation in resi
the current errors may be related to previous errors,
 violating the independence of errors assumption in Linear Regression.
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#3. Explain why residuals should be plotted over time.
Residuals should be plotted over time to detect autocorrelation and repeating
 errors are independent or related to previous time periods, ensuring the inde
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# 4.Describe how seasonality or omitted time-related variables could cause thi
Seasonality: Sales may increase or decrease during specific days, months, fest
Omitted Time-related Variables: Factors like weekends, holidays, seasons, or p
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# Suggest appropriate improvements to the model.
Include time-related features such as day, month, weekends, and holidays.
Add lagged variables, such as previous day's sales.
Use time-series models like ARIMA or SARIMA to handle autocorrelation and seas
Check residuals again using a residual plot and Durbin-Watson test.
```